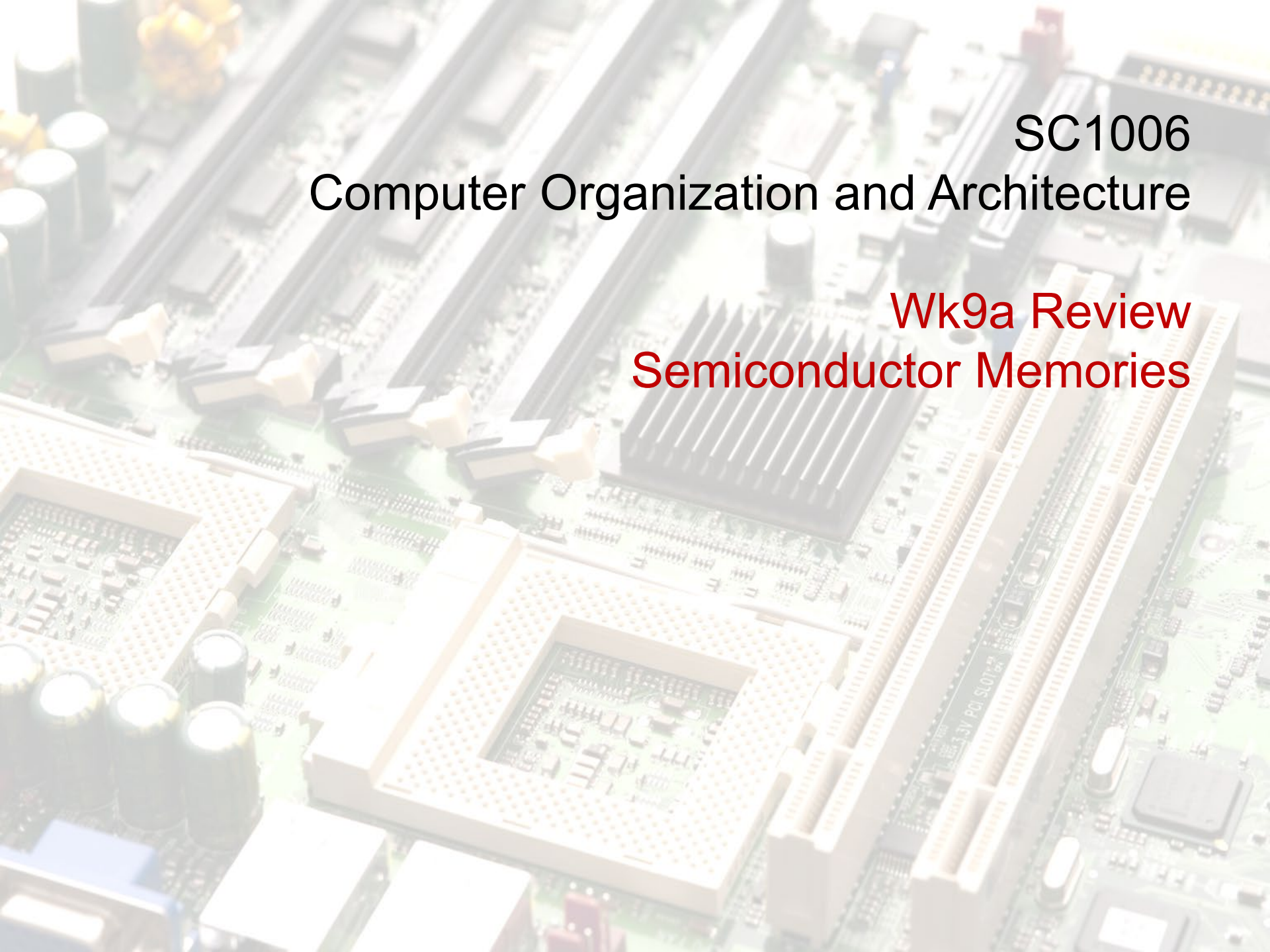




SC1006

Computer Organization and Architecture

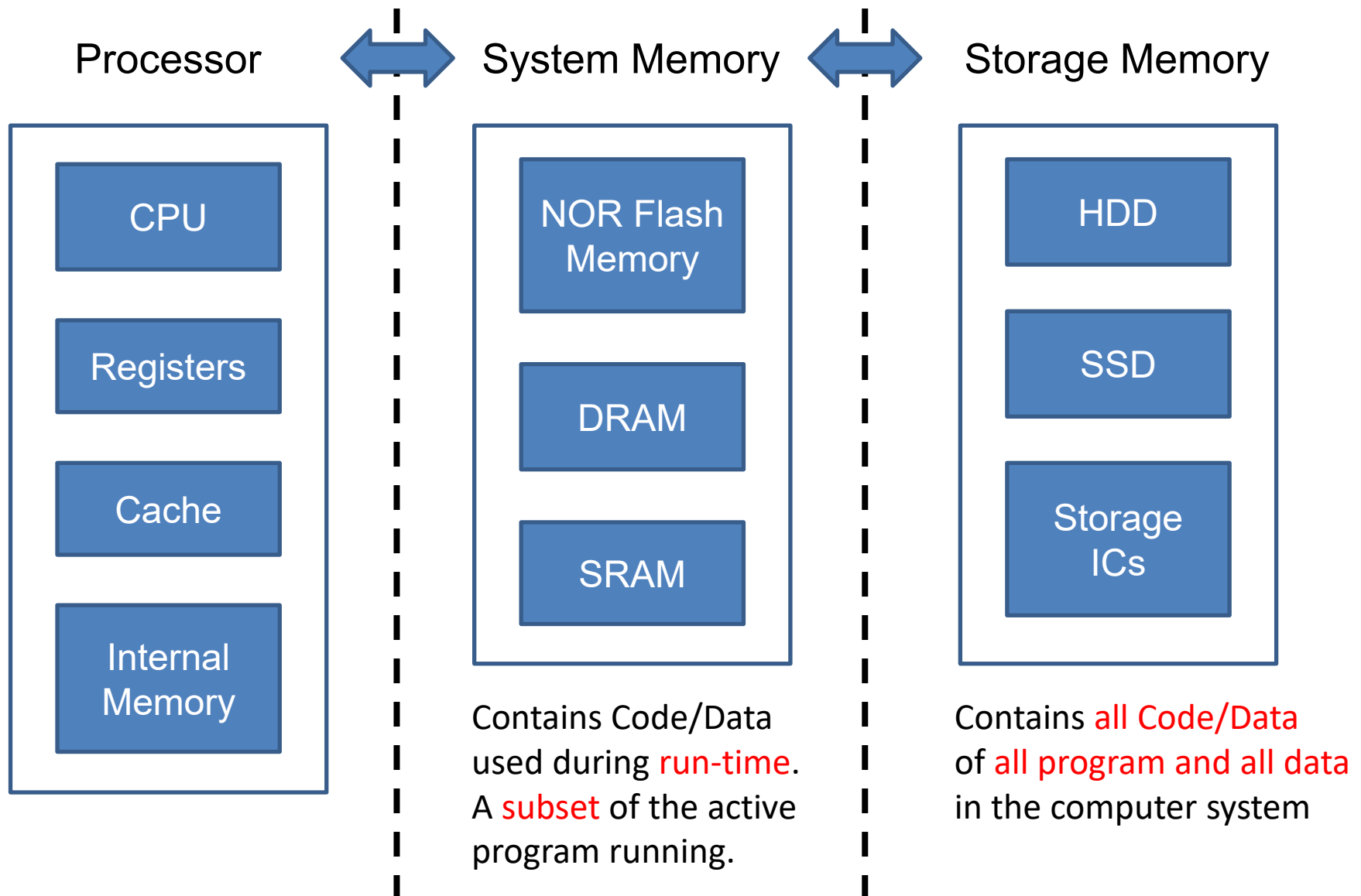
Wk9a Review

Semiconductor Memories

An important topic in a company project

- Hardware Design
- Software Implementation
- Co- Design of Hardware & Software

Computer Memory (Functional View)

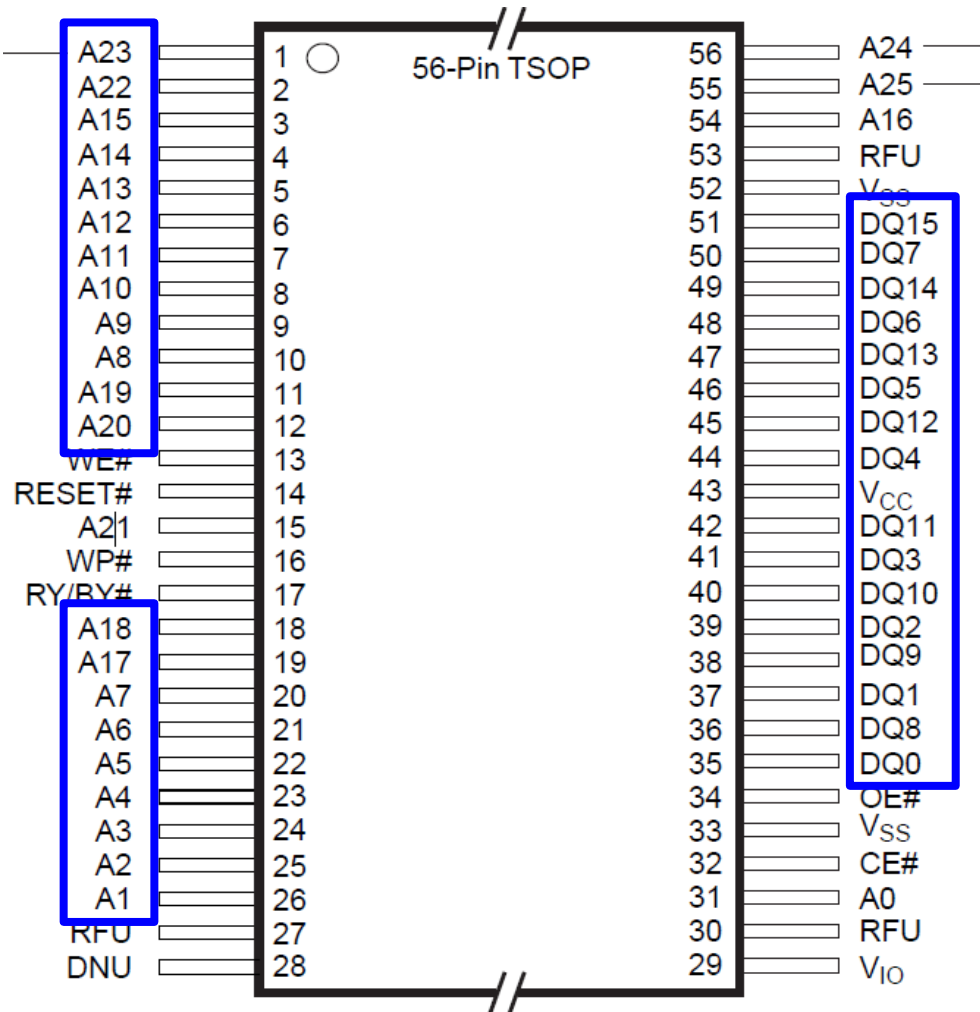


System vs Storage Memory

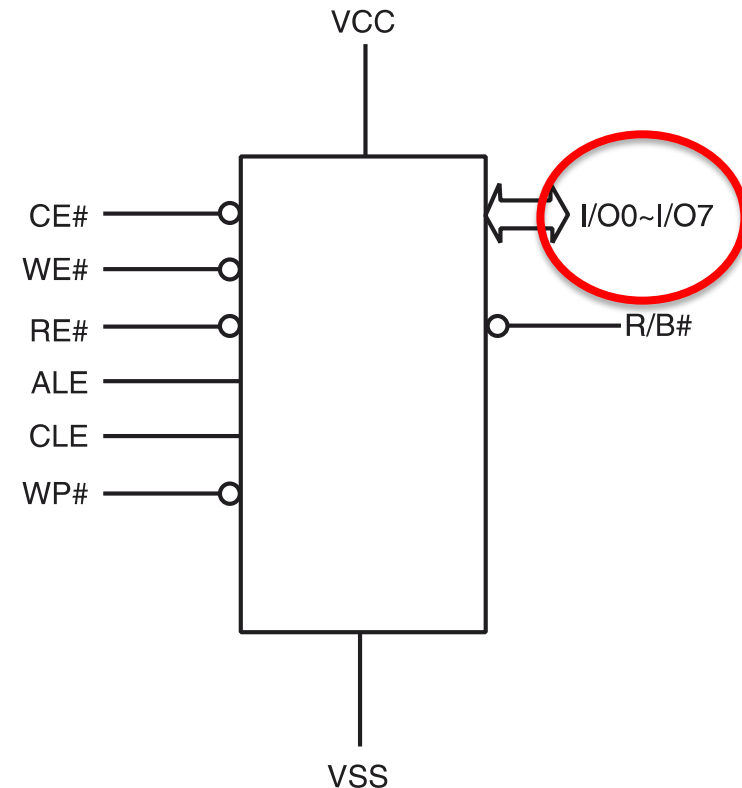
- System Memory
 - Used to store runtime program; code is executed directly from the system memory
 - Use NOR Flash, SRAM, DRAM
 - Support XIP
 - DRAM technically speaking does not support XIP by itself but processors that has a DRAM interface will have a DRAM controller that handle the necessary translation to allow execution of code directly from the DRAM.
- Storage memory
 - Used to store all program and data in a computer system
 - Cannot run code directly from storage memory, needs to be transferred to system memory before code execution
 - All memory types can be used as storage memory

NOR and NAND Flash chip pin-out

NOR Flash



NAND Flash



Thinking questions toward innovation...

A open question: How does memory play a crucial role in the development and operation of AI systems?

